

Zigbee Router / Repeater Setup Guide

SONOFF ZBDongle-E V2 as Mesh Range Extender

Awesom Living — Pilot Home Deployment

Field	Value
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Estimated Time	15–20 minutes (excluding re-pairing sensors)

1. Overview

The Awesom Living pilot home uses a SONOFF ZBDongle-E V2 plugged into the Raspberry Pi as the Zigbee coordinator. All Zigbee sensors (motion, presence, door/window, emergency button) communicate directly with this coordinator. In larger homes or homes with thick walls, sensors placed far from the coordinator may experience weak signal (low linkquality), leading to delayed or missed events.

A Zigbee router (repeater) extends the mesh by relaying messages between distant sensors and the coordinator. A spare SONOFF ZBDongle-E V2 can be repurposed as a dedicated router — it requires only USB power (any phone charger) and no connection to the Pi.

1.1 Zigbee Range — How Far Can Sensors Reach?

Zigbee operates on the 2.4 GHz frequency band. Its effective range depends heavily on the environment — walls, furniture, and other 2.4 GHz devices (WiFi routers, microwaves) all reduce signal strength. The table below shows typical ranges for the SONOFF ZBDongle-E V2 coordinator:

Environment	Effective Range	Notes
Open air (no obstacles)	30–50 meters	Theoretical maximum; rarely applies indoors
Single room, line of sight	15–20 meters	Strong signal; repeater NOT needed
Through 1 drywall / wooden wall	10–15 meters	Adequate signal; repeater usually NOT needed
Through 1 brick / concrete wall	7–10 meters	Signal weakens noticeably; monitor linkquality
Through 2+ brick / concrete walls	3–6 meters	Repeater REQUIRED — signal degrades severely

Environment	Effective Range	Notes
Through reinforced concrete / metal door	1–3 meters	Repeater REQUIRED — near-total signal block

1.2 When a Repeater is NOT Required

You do NOT need a repeater in these scenarios:

- Small apartment or flat (1–2 BHK) where all rooms are within 10 meters of the Pi
- Sensors are in the same room or adjacent room as the coordinator with only drywall / wooden partitions
- All sensors show linkquality above 100 in the Zigbee2MQTT dashboard
- Events (motion, door open/close) arrive instantly with no delays or missed events
- Home has an open floor plan without thick walls between the Pi and sensors

1.3 When a Repeater IS Required

Add a repeater when any of these conditions apply:

- Any sensor's linkquality is below 80 in the Zigbee2MQTT dashboard
- Sensors are more than 10 meters from the coordinator with brick or concrete walls in between
- The home is a 3+ BHK flat or an independent house with rooms spread across a larger area
- Events (motion, door open/close) arrive late (more than 2–3 seconds delay) or are missed entirely
- The coordinator (Pi) is in one corner of the home and sensors are placed in rooms on the opposite side
- There are 2 or more concrete / brick walls between the coordinator and the farthest sensor
- The home has metal doors, metal window frames, or reinforced concrete ceilings between floors

1.4 Understanding Linkquality Values

Linkquality (LQI) is reported by Zigbee2MQTT on a scale of 0–255. Higher is better. Use it as the primary indicator for whether a sensor needs a repeater:

Linkquality Range	Signal Strength	Action
200–255	Excellent	No action needed
120–199	Good	Acceptable — monitor for drops
80–119	Fair	Borderline — consider a repeater if events are delayed
40–79	Weak	Repeater REQUIRED — frequent packet loss likely
0–39	Critical	Repeater REQUIRED — sensor may appear

Linkquality Range	Signal Strength	Action
		offline

1.5 What You Need

Item	Purpose	Notes
SONOFF ZBDongle-E V2 (spare)	The dongle to flash as a router	Must NOT be the one plugged into the Pi
Windows / Mac / Linux laptop	Run the web flasher	Chrome or Edge browser required
USB-A port or USB-A extension cable	Connect dongle to laptop for flashing	Also used for final deployment
USB charger (5V, any amperage)	Power the dongle after flashing	Any phone charger works
USB-A extension cable (optional)	Better placement flexibility	Lets you position the repeater optimally

2. Flash the Dongle with Router Firmware

 NEVER flash the coordinator dongle (the one plugged into the Pi). The spare dongle is the one you flash. If in doubt, unplug the Pi's dongle, label it "COORDINATOR — DO NOT FLASH", and work only with the other one.

2.1 Identify Your Dongle Chip

The SONOFF ZBDongle-E V2 uses the Silicon Labs EFR32MG21 chip. This is important because the firmware file and flasher tool are chip-specific. The older ZBDongle-P uses a different chip (CC2652P) and requires different firmware.

Confirm by checking the label on the dongle PCB or the product box — it should say "ZBDongle-E" or "EFR32MG21".

2.2 Download Router Firmware

Step 1: Go to the firmware repository

Open: <https://github.com/Koenkk/Z-Stack-firmware/tree/master/router/Silabs/EFR32MG21>
(or search "Koenkk Z-Stack-firmware silabs router" on GitHub)

Step 2: Download the correct .gbl file

File name: router_EFR32MG21_d-1-1_115200.gbl

 Do NOT download "coordinator" firmware — that is for the Pi dongle. You need the "router" firmware.

2.3 Flash Using the Web Flasher

Step 1: Open the web flasher

Go to: <https://darkxst.github.io/silabs-firmware-builder/>
(Works in Chrome and Edge. Firefox may not support WebSerial.)

Step 2: Plug the spare dongle into your laptop USB port

If the dongle previously had coordinator firmware, it may be detected as a serial device. That is normal.

Step 3: Connect to the dongle

Click "Connect" in the web flasher. A browser dialog will show available serial ports. Select the one labeled "Silicon Labs" or "SONOFF" (usually the only new entry).

Step 4: Upload the firmware file

Click "Choose file" and select the router_EFR32MG21_d-1-1_115200.gbl file you downloaded.

Step 5: Flash

Click "Flash" / "Upload". The process takes 30–60 seconds. The progress bar should reach 100%.

Do not unplug the dongle during flashing.

Step 6: Verify

The flasher should show "Firmware flashed successfully" or similar. Unplug the dongle from the laptop.

 If the web flasher fails to connect, try: (a) a different USB port, (b) a different browser (Chrome is most reliable), (c) close any other serial monitor software (PuTTY, Z2M, etc.) that might be holding the port.

3. Deploy the Repeater

3.1 Choose the Placement

The repeater should be placed roughly halfway between the coordinator (Raspberry Pi) and the farthest sensors. Ideal placement:

- Central hallway or corridor connecting rooms
- Elevated position (shelf height, 1–1.5 meters) — avoid floor level
- Line of sight to both the Pi and the distant sensors if possible
- Near a USB power outlet (any phone charger or USB wall adapter)
- Away from large metal objects, microwave ovens, and WiFi routers (2.4 GHz interference)

3.2 Power On

Step 1: Plug the flashed dongle into a USB charger

Use any standard 5V USB charger. An extension cable helps with placement.

The dongle's green LED should start blinking — this means it is powered and looking for a Zigbee network to join.

3.3 Join the Zigbee Network

Step 1: Open Zigbee2MQTT dashboard

URL: `http://192.168.50.106:8080` (or your Pi's IP on port 8080)

Step 2: Enable Permit Join

Click the "Permit join" button at the bottom of the Z2M dashboard. This allows new devices to join for a limited time (usually 4 minutes).

Step 3: Wait for the dongle to appear

Within 30–60 seconds, the router dongle should appear as a new device in Z2M. It will show as type "Router" (not "End Device").

The device count should increase by 1, and "Router: 1" should appear in the device summary.

Step 4: Rename the device

Click the new device → Settings icon → Rename it to "Repeater" (or a location-specific name like "hallway_repeater").

Step 5: Verify router status

In the device details, confirm:

Type:	Router
Power source:	DC source (mains)

Link quality: Should show a value (e.g. 200+)

3.4 Register in Backend (Admin Panel)

If the Awesom Living admin panel is used for device management, map the repeater there as well:

- Device name: Repeater
- Type: Zigbee
- Sensor type: Router
- Status: Mapped
- The repeater does not generate sensor events — it only relays messages

4. Re-pair Sensors Through the Repeater

⚠ CRITICAL: Battery-powered Zigbee sensors ("sleepy end devices") do NOT automatically discover new routers. They pick their parent during pairing and stick with it permanently. To benefit from the repeater, each distant sensor must be removed and re-paired.

4.1 Priority Order

Re-pair sensors starting with the weakest signal (lowest linkquality):

Priority	Sensor	Current LQ	Action
1 (Highest)	motion_2 Sensor Master	96	Re-pair first
2	Motion Sensor Master	124	Re-pair next
3	Door Sensor	132	Re-pair if still weak
4	window Sensor Master	132	Re-pair if still weak
5 (Optional)	Emergency SOS	232	Signal already strong — skip unless issues

4.2 Re-pairing Procedure (Per Sensor)

Step 1: Remove the sensor from Z2M

In Z2M dashboard → click the sensor → click the Delete icon (trash can) → confirm.
If normal removal fails, use "Force remove".

Step 2: Enable Permit Join in Z2M

Click "Permit join" — you have ~4 minutes to complete pairing.

Step 3: Put the sensor in pairing mode

Method varies by sensor type:

Sensor Type	Pairing Method
Motion Sensor (PIR)	Hold the reset button (small hole on back) for 5+ seconds until LED blinks rapidly
Door/Window Sensor	Hold the reset button for 5+ seconds until LED blinks
Emergency Button	Hold the button for 10 seconds until LED flashes

Step 4: Wait for it to join

The sensor should appear in Z2M within 30–60 seconds as a new device.

Step 5: Rename the sensor

Restore its original friendly name (e.g. "Motion Sensor Master", "Door Sensor").

Step 6: Check the routing

In the Z2M device details → "About" tab, look at the network route. If the repeater is closer, the sensor should now route through it. The linkquality should be noticeably higher than before.

Step 7: Verify sensor events

Trigger the sensor (wave hand for motion, open/close door) and confirm events appear in Z2M's live log and in the Awesom Living app.

 After re-pairing, verify that the sensor's Zigbee2MQTT friendly name matches the name expected by Node-RED flows and the backend. Mismatched names mean the backend won't receive events from that sensor.

5. Verification Checklist

Check	Expected Result	Pass?
Repeater visible in Z2M	Shows as "Router" type with linkquality > 0	☐
Z2M device count correct	Total devices = sensors + 1 (repeater)	☐
Repeater powered independently	Plugged into USB charger, NOT the Pi	☐
Re-paired sensors show higher LQ	linkquality improved vs. before	☐
Motion sensor triggers events	Wave hand → occupancy:true appears in Z2M log	☐
Door sensor triggers events	Open door → contact:false appears in Z2M log	☐
Events reach the app	Dashboard updates within 2–3 seconds of sensor trigger	☐
Push notifications work	Notification received on phone (dev build only)	☐

6. Troubleshooting

6.1 Repeater won't join the network

- Ensure "Permit join" is enabled in Z2M before powering on the dongle
- Power-cycle the dongle (unplug, wait 5 seconds, plug back in)
- Ensure the dongle has router firmware, not coordinator firmware
- Check the Z2M coordinator is running (Z2M dashboard should be reachable)

6.2 Sensor signal not improving after re-pair

- Confirm the sensor actually routes through the repeater (Z2M → device → "About" tab → network route)
- Move the repeater closer to the weak sensor
- Ensure no large metal objects or microwaves between repeater and sensor
- Try re-pairing with the sensor physically next to the repeater, then move it to final position

6.3 Sensor events not reaching backend after re-pair

- Check that the sensor's friendly name in Z2M matches exactly what Node-RED expects
- Check Node-RED flows at <http://<Pi-IP>:1880> — ensure MQTT topics match
- Check backend logs for incoming webhook events
- In Z2M, go to the sensor and click "Reconfigure" to re-bind clusters

7. Quick Reference

Item	Value
Coordinator dongle	SONOFF ZBDongle-E V2 → plugged into Pi USB 3.0
Router dongle	SONOFF ZBDongle-E V2 (spare) → plugged into USB charger
Router firmware	router_EFR32MG21_d-1-1_115200.gbl
Firmware repo	github.com/Koenkk/Z-Stack-firmware/tree/master/router/Silabs/EFR32MG21
Web flasher	darkxst.github.io/silabs-firmware-builder/
Z2M dashboard	<a href="http://<Pi-IP>:8080">http://<Pi-IP>:8080
Node-RED dashboard	<a href="http://<Pi-IP>:1880">http://<Pi-IP>:1880
Pi IP (office)	192.168.50.106